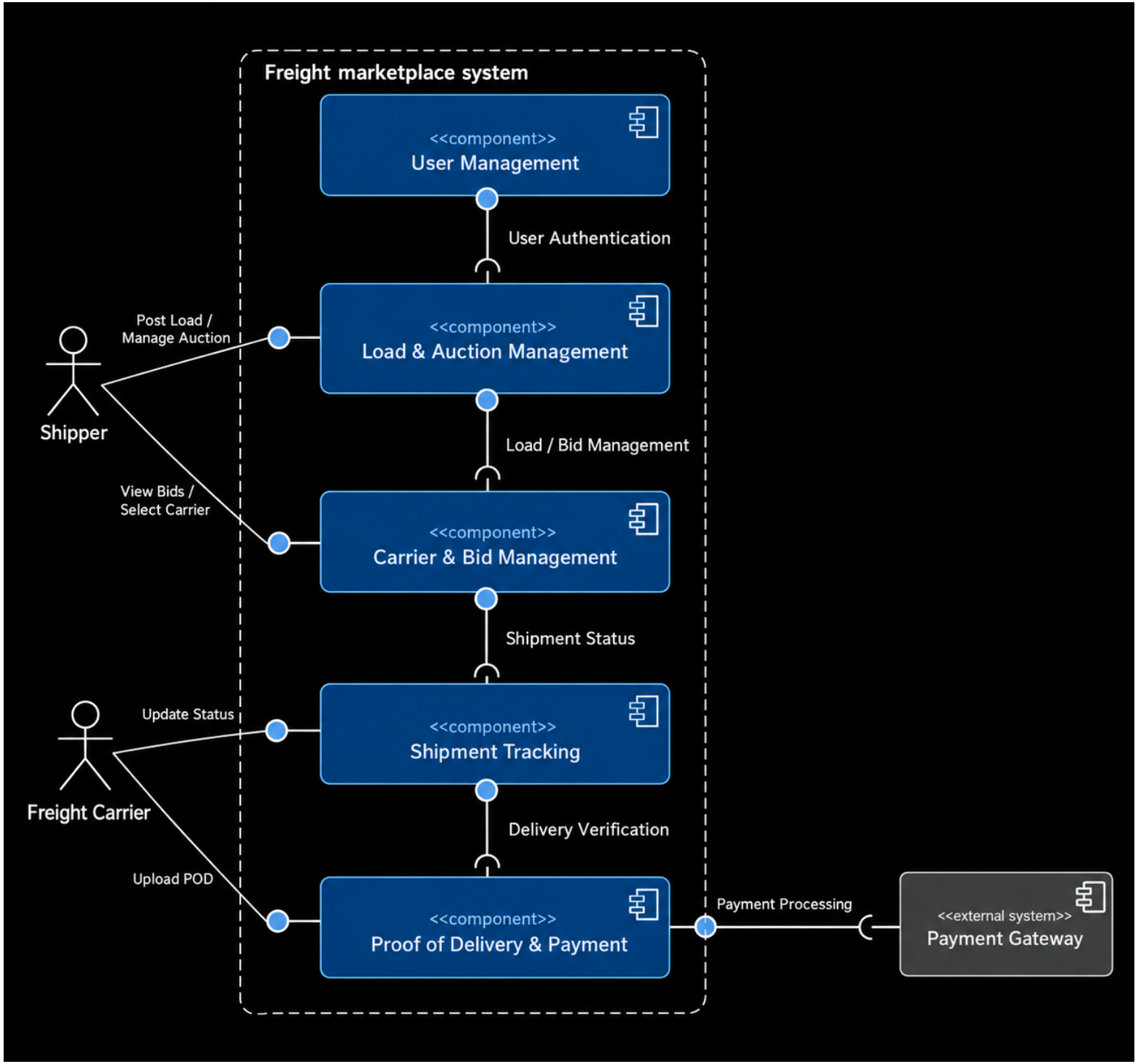

Freight marketplace system
<<component>>
User Management
User Authentication
<<component>>
Load & Auction Management
Post Load /
Manage Auction
Shipper
Load / Bid Management
View Bids /
Select Carrier
<<component>>
Carrier & Bid Management
Shipment Status
Update Status
<<component>>
Shipment Tracking
Freight Carrier
Delivery Verification
Upload POD
<<component>>
Proof of Delivery & Payment
Payment Processing
<<external system>>
Payment Gateway

FreightMatch – Architecture Selection & Justification

Architecture Selection

We chose Microservices Architecture for the Inter-City Freight Load Matching Marketplace (FreightMatch). The system supports multiple independent business functions such as freight-load posting and auction management, carrier and bid management, shipment tracking, and proof-of-delivery and payment processing. A microservices architecture allows these functions to be separated into focused, independently manageable components.

Reason 1 – Independent Scalability

Freight bidding, shipment tracking, and payment processing can experience different workloads. Microservices allow individual services to be scaled according to demand instead of scaling the entire marketplace together. This is useful when many carriers are submitting bids or when several shipments are being tracked simultaneously.

Reason 2 – Fault Isolation and Independent Deployment

The major functions of FreightMatch can be developed, updated, and maintained independently. A problem or change in one service, such as payment processing, can be isolated from other services such as load management or shipment tracking. This improves maintainability and reduces the impact of failures or updates on the rest of the marketplace.

Security Advantage

Security-sensitive functions such as carrier verification and payment processing can be isolated into dedicated services with their own access controls. This supports controlled communication between services and helps protect sensitive shipment, user, and payment-related operations.

Performance Benefit

Independent services can process requests concurrently and can be scaled separately when demand increases. For example, bid-submission traffic can be handled independently from shipment-tracking and payment-processing requests, helping the marketplace remain responsive under higher workloads.